

PROGRESS REPORT

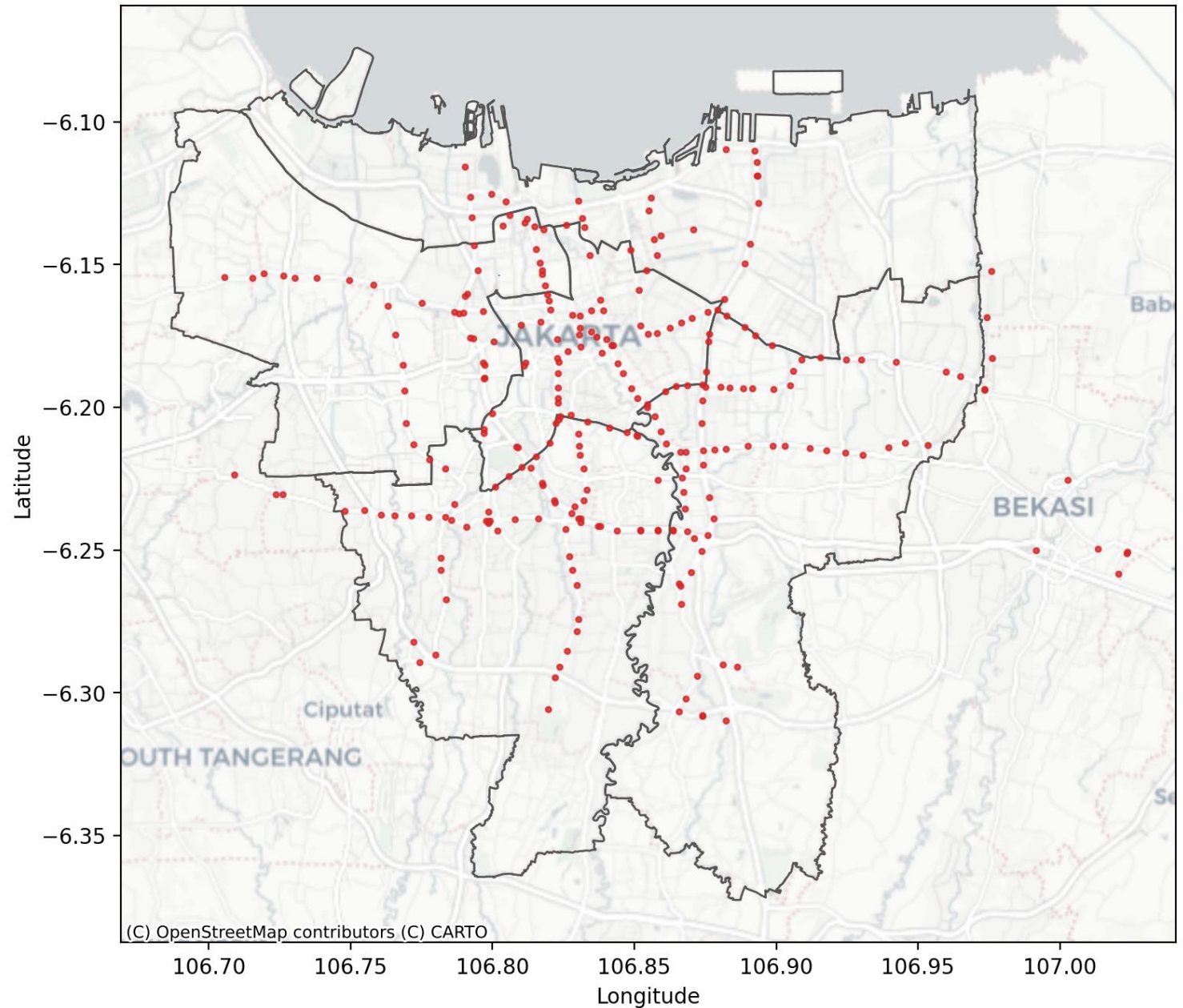
Public Transit User Dynamics During Boarding and Alighting: Video Data Analysis of Jakarta Public Transport Network

Damar Pratita Dewayanto
Supervisor: Prof. Makoto Chikaraishi

Video Data Availability

TransJakarta - Jakarta Integrated Bus Transit System

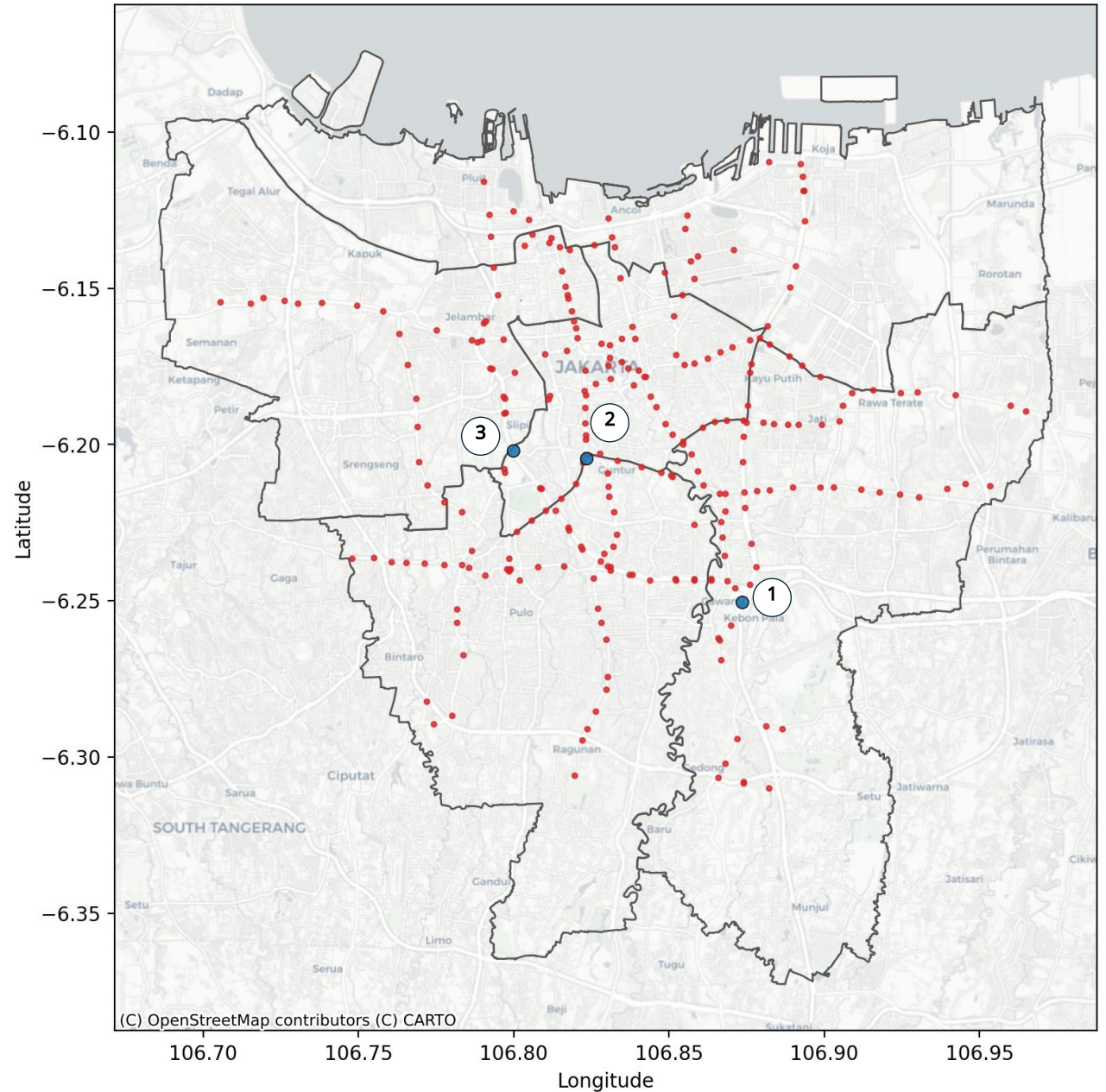
- TransJakarta – Jakarta Integrated Bus Transit System that serves not only Jakarta but suburban outside Jakarta administration
- Distribution of bus station – a station with dedicated platform with multiple docking bays to serve multiple route mostly for BRT services– across the network



Video Data Availability

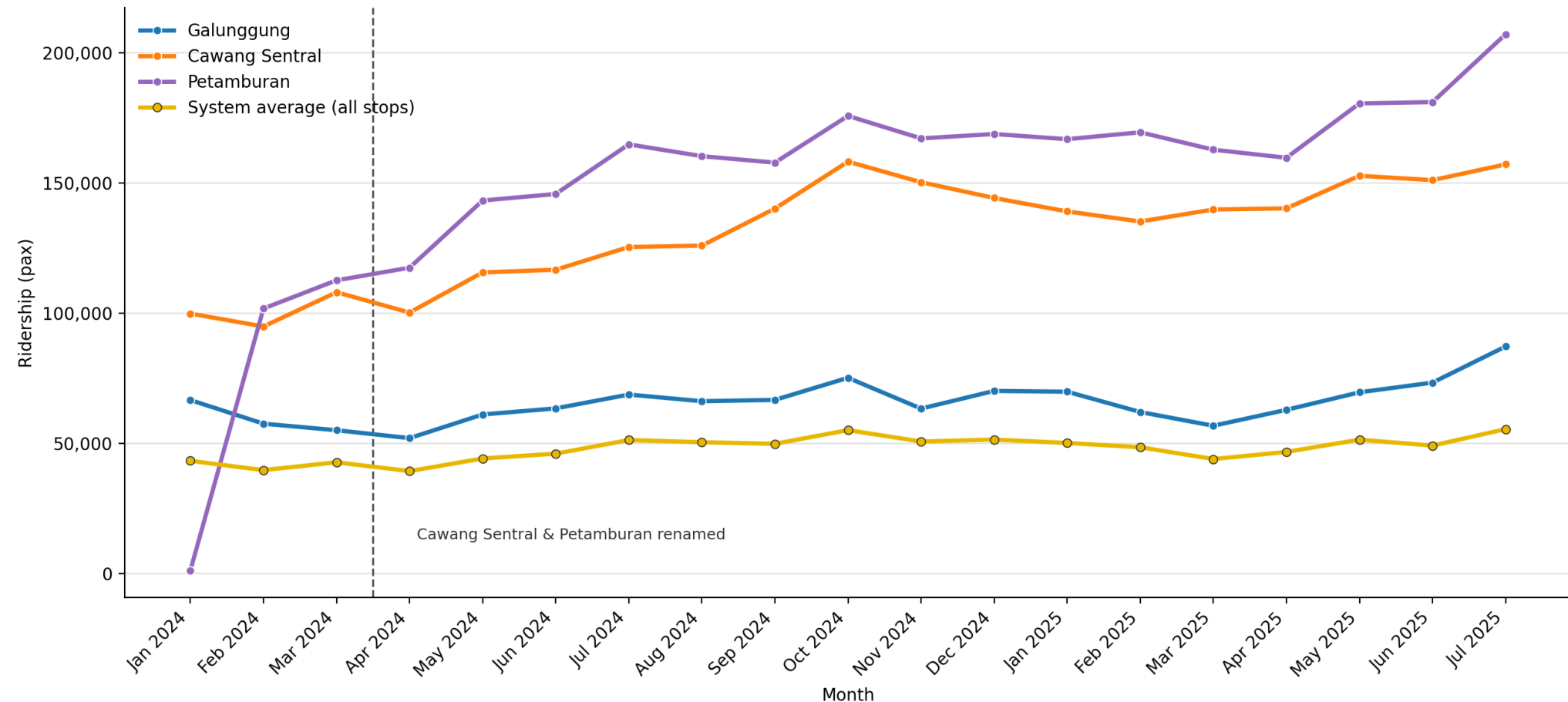
TransJakarta - Jakarta Integrated Bus Transit System

- Among all stations, the available video data covers following stations:
 1. Cawang Sentral
 2. Galunggung
 3. Petamburan



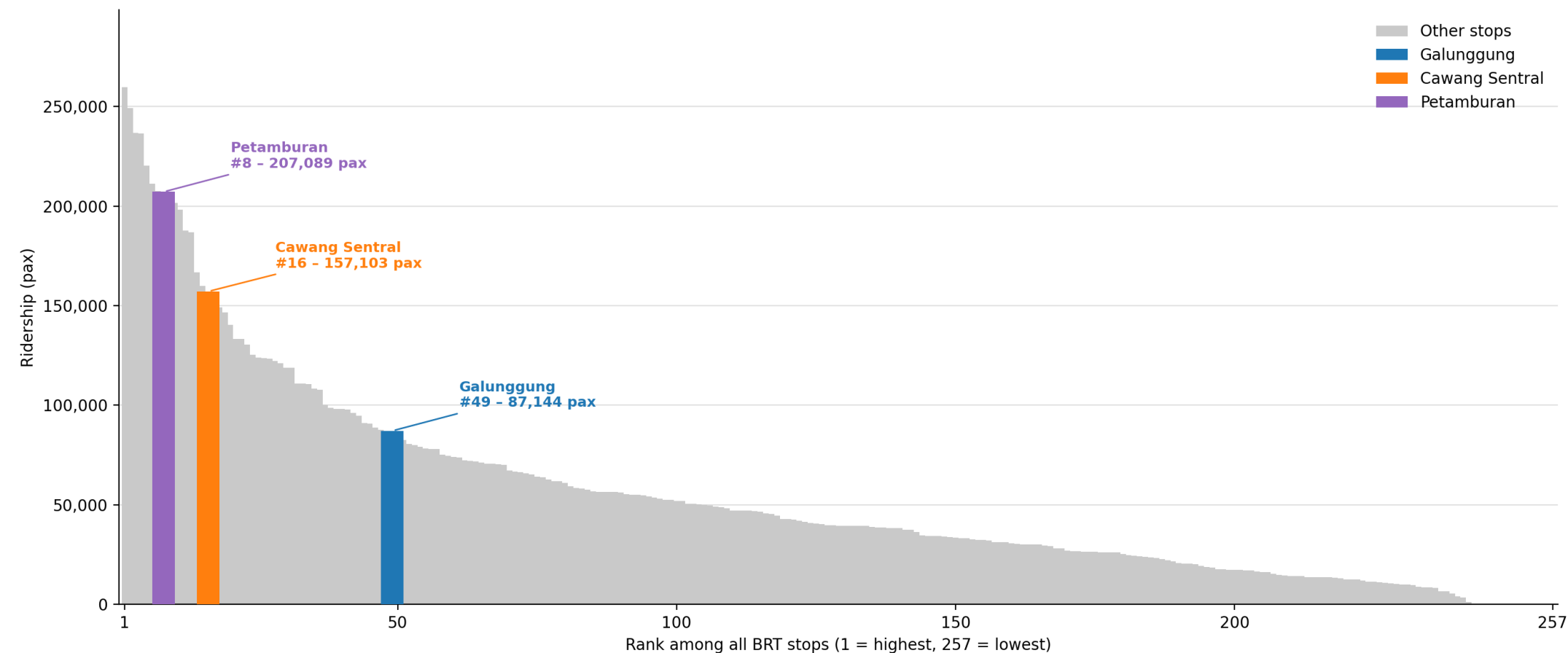
Video Data Availability

Three Bus Station – Monthly Passenger Jan 2024 to July 2025



Video Data Availability

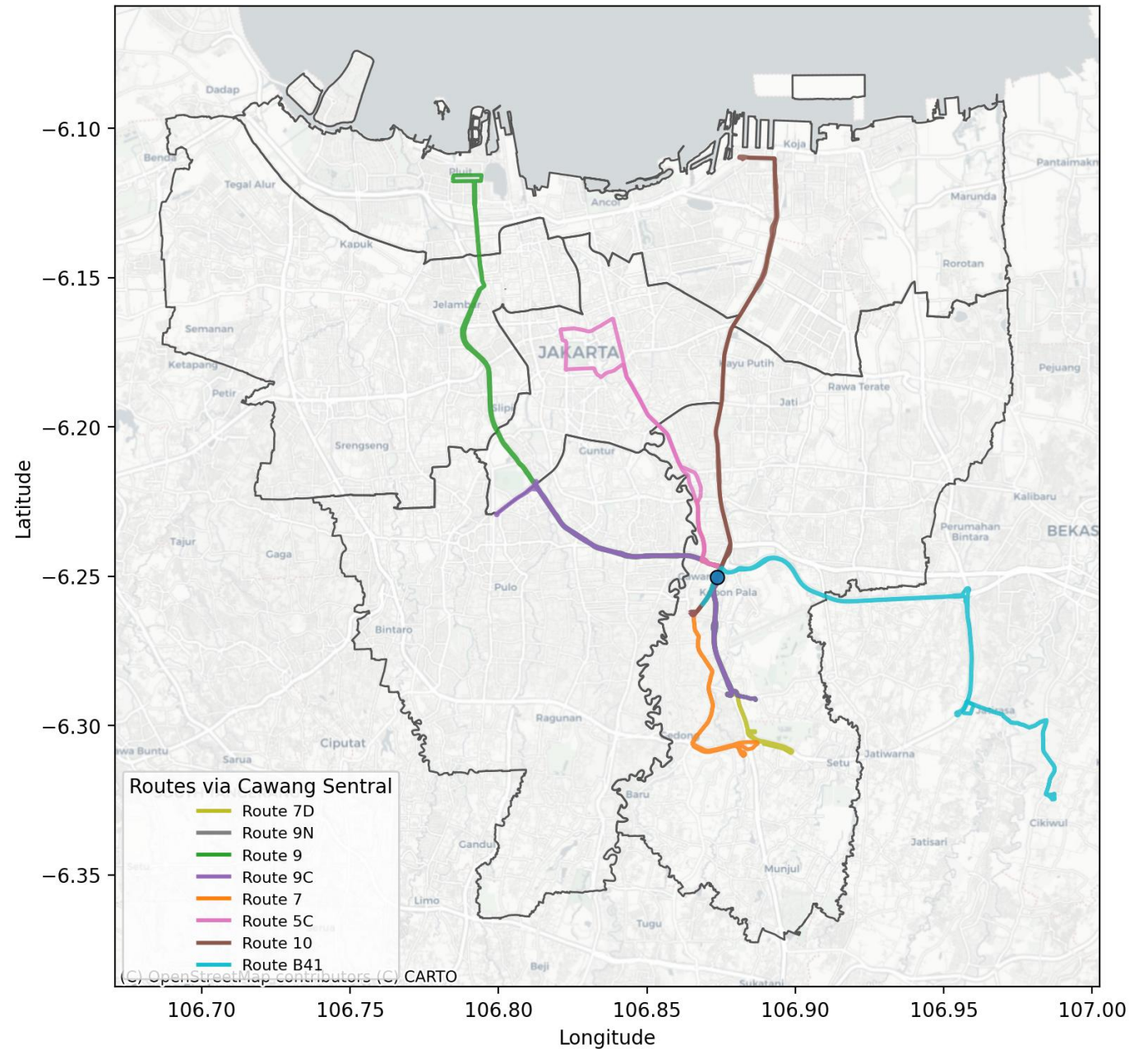
Three Bus Station Among All Stations – Number of Passenger July 2025



Video Data Availability

Cawang Sentral – Network Coverage

- Network coverage at Cawang Sentral: an 8-route, 4-direction interchange including direct route to/from Bekasi (blue line)



Video Data Availability

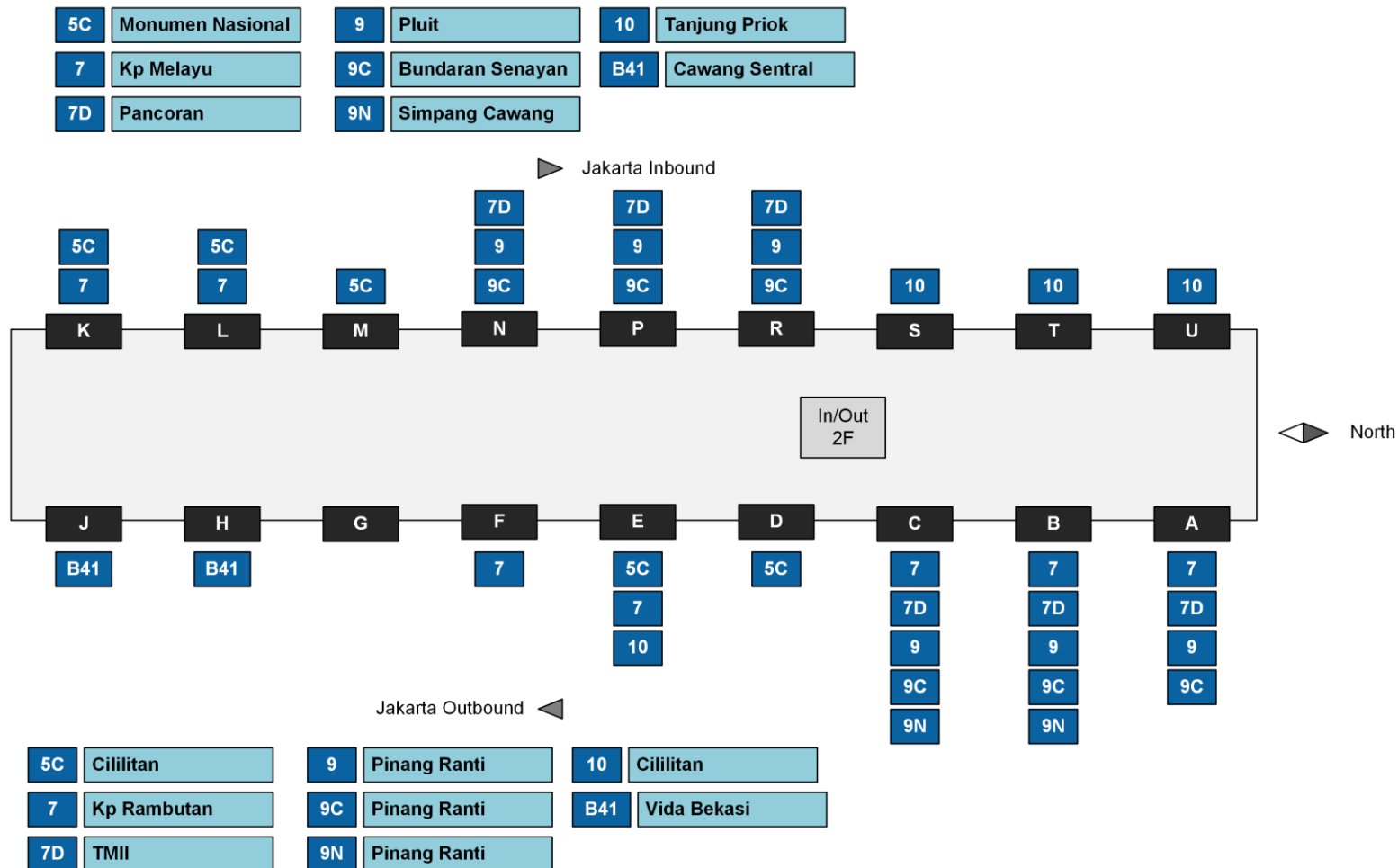
Cawang Sentral – Land Use

- Cawang Sentral Bus Station sits at the core of Jakarta's largest toll interchange where other BRT and LRT Jabodebek converge nearby, but at two separate nodes ~450m apart.



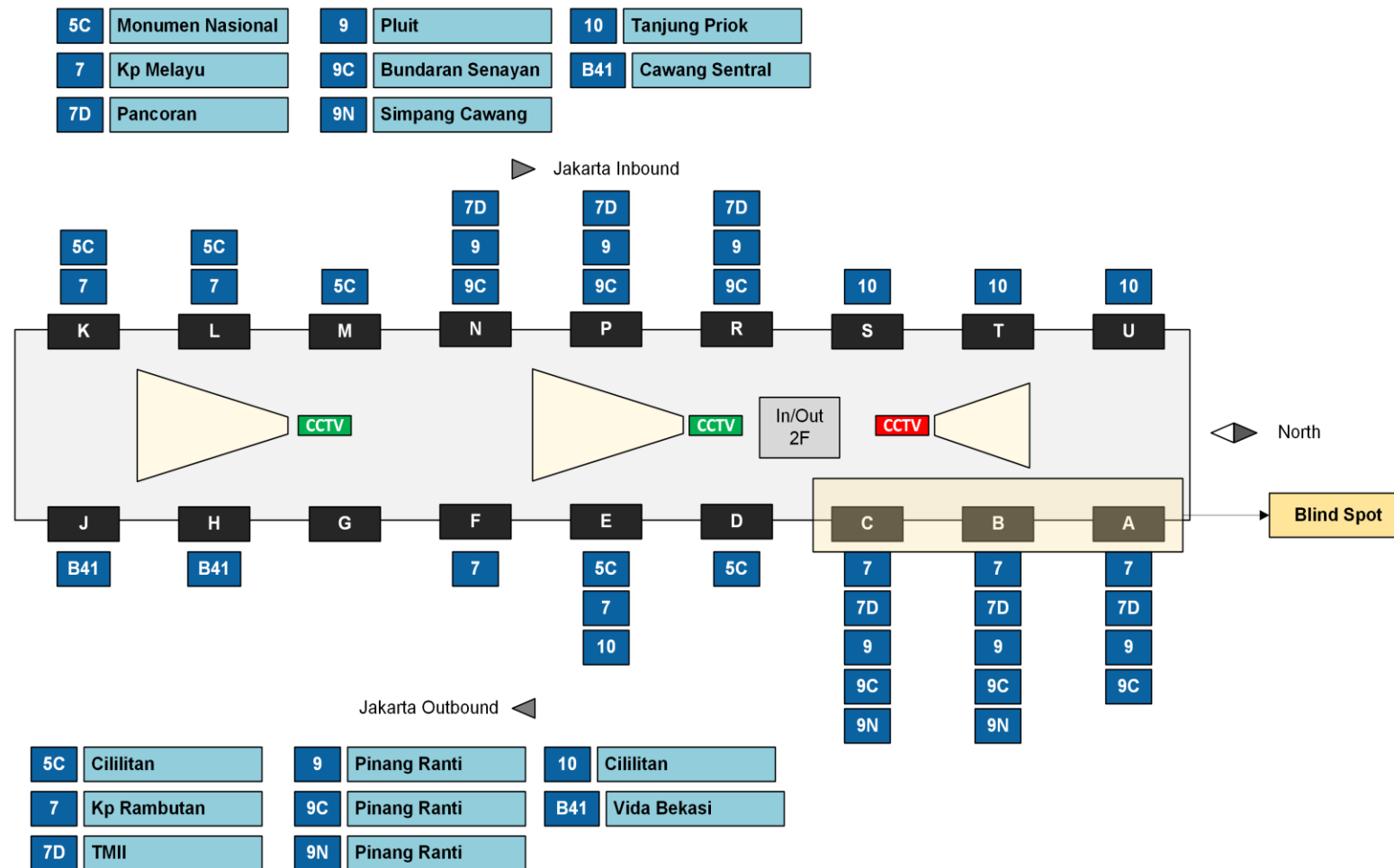
Video Data Availability

Cawang Sentral – Station Layout



Video Data Availability

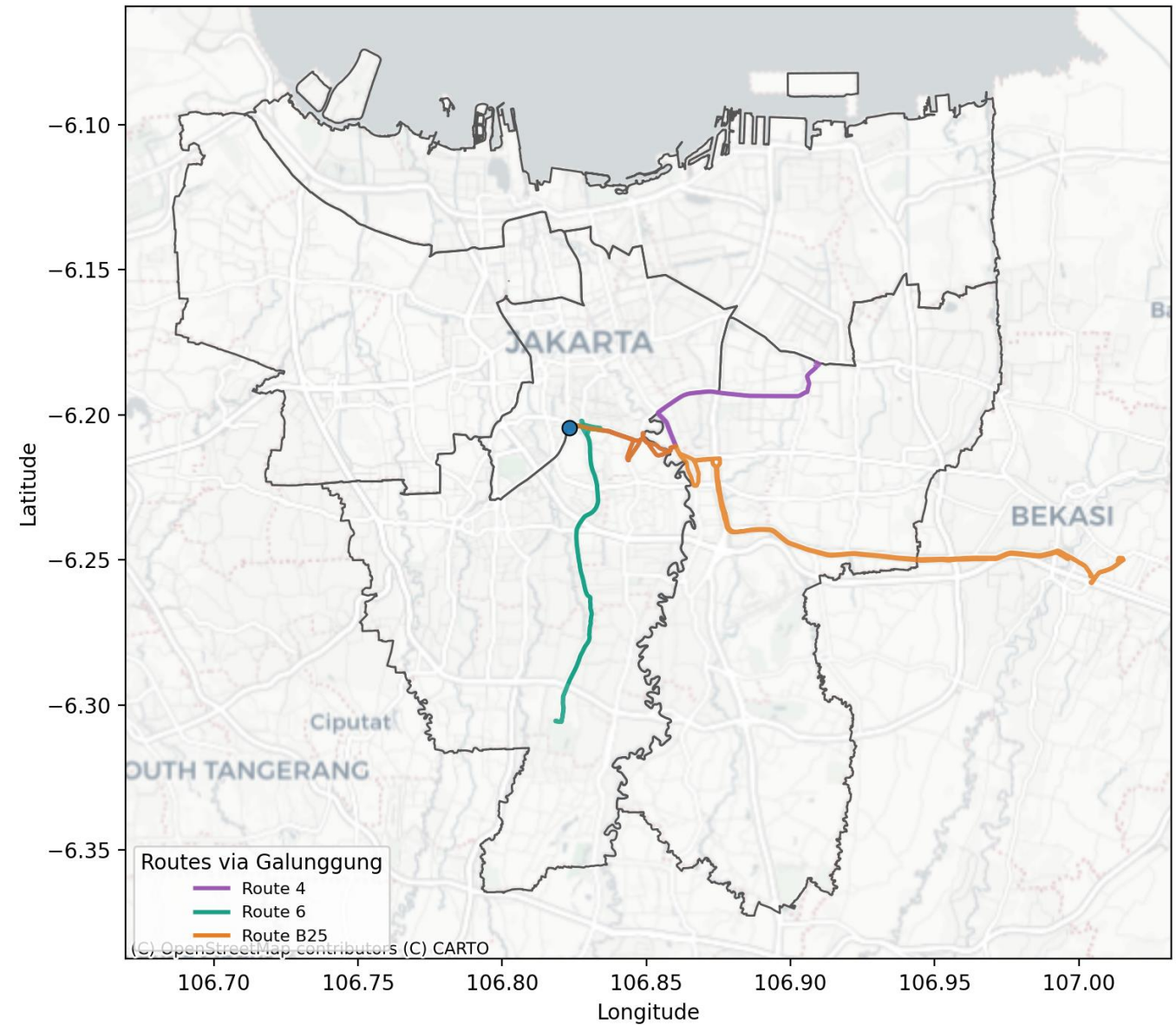
Cawang Sentral – Camera Spot



Video Data Availability

Galunggung – Network Coverage

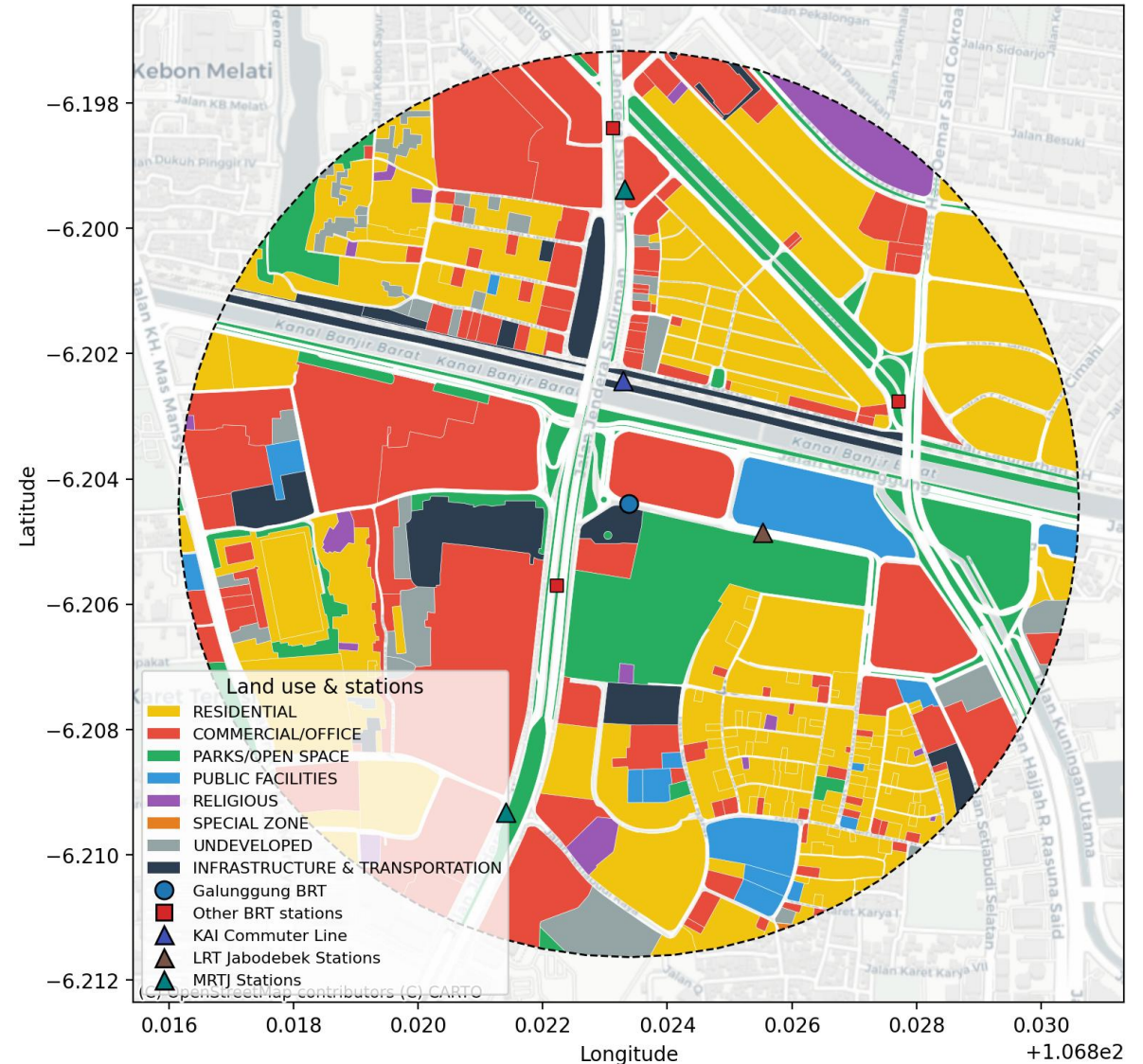
- Network coverage Galunggung: a 3-route bus station, with reach extending east to Bekasi via Route B25 (orange line)



Video Data Availability

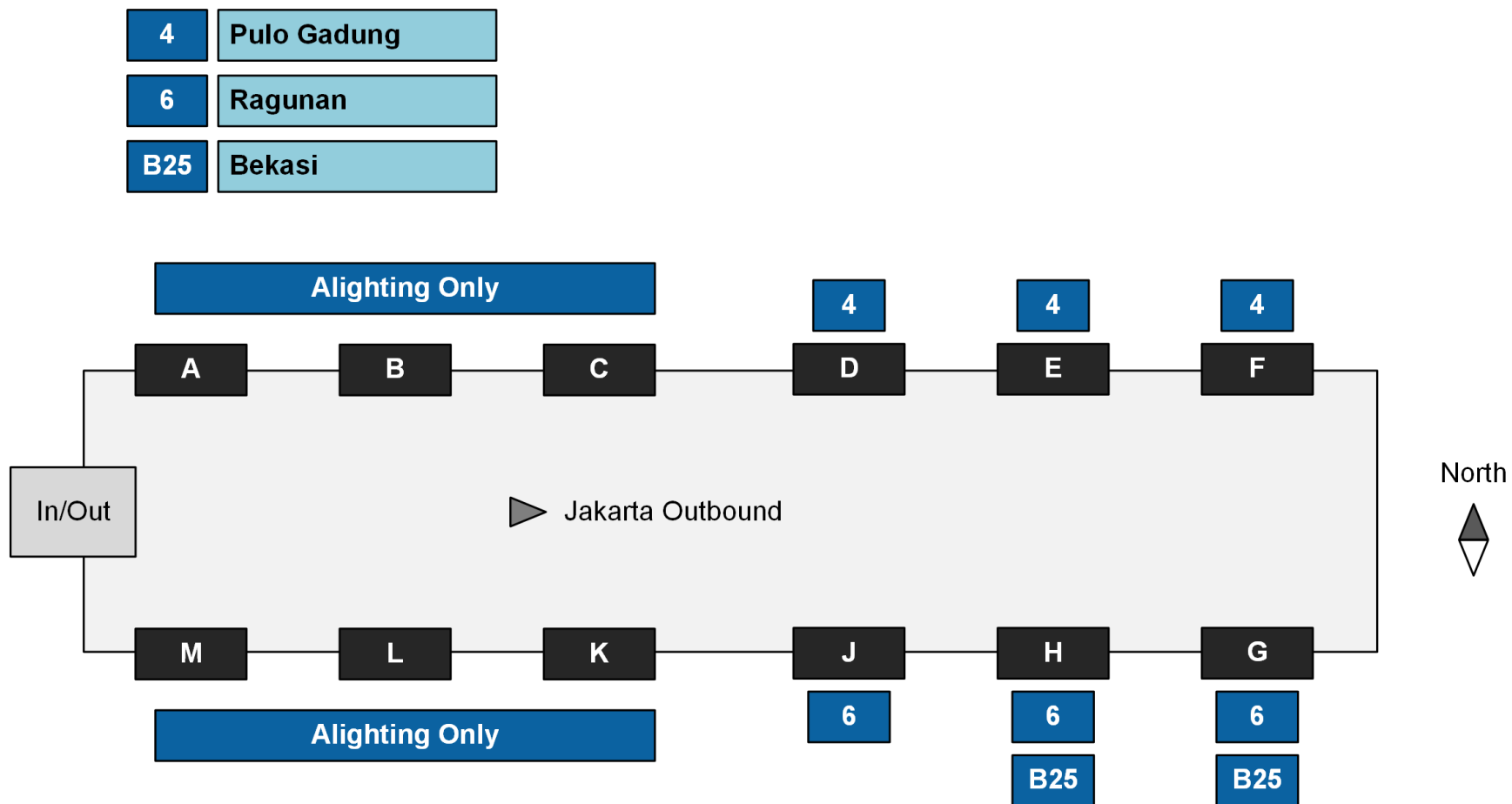
Galunggung – Land Use

- Galunggung Bus Station is located around major public transport station (MRT Jakarta, LRT Jabodebek, KAI Commuter Line, and another BRT line) serving as important transit node
- Galunggung Bus Station is located around Jakarta's central business corridor (Sudirman – Thamrin)



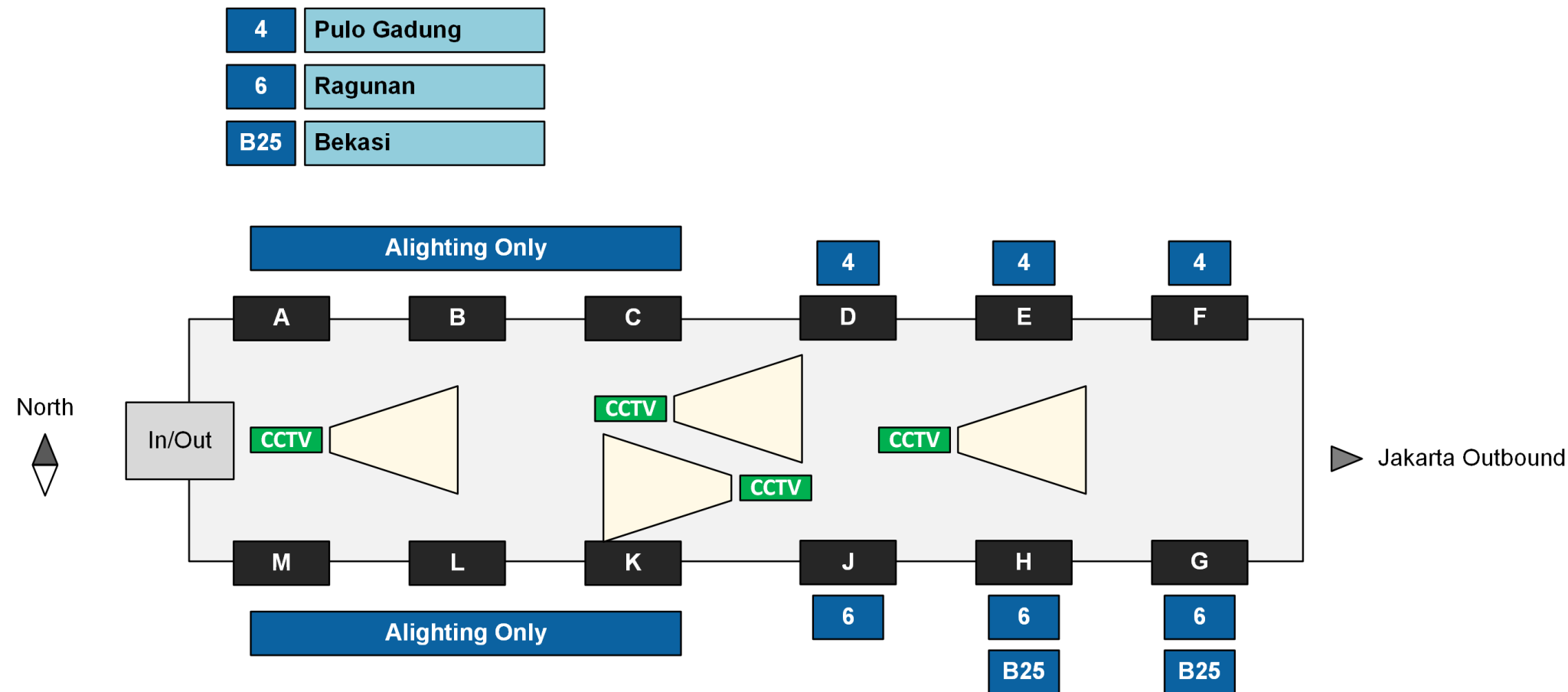
Video Data Availability

Galunggung – Station Layout



Video Data Availability

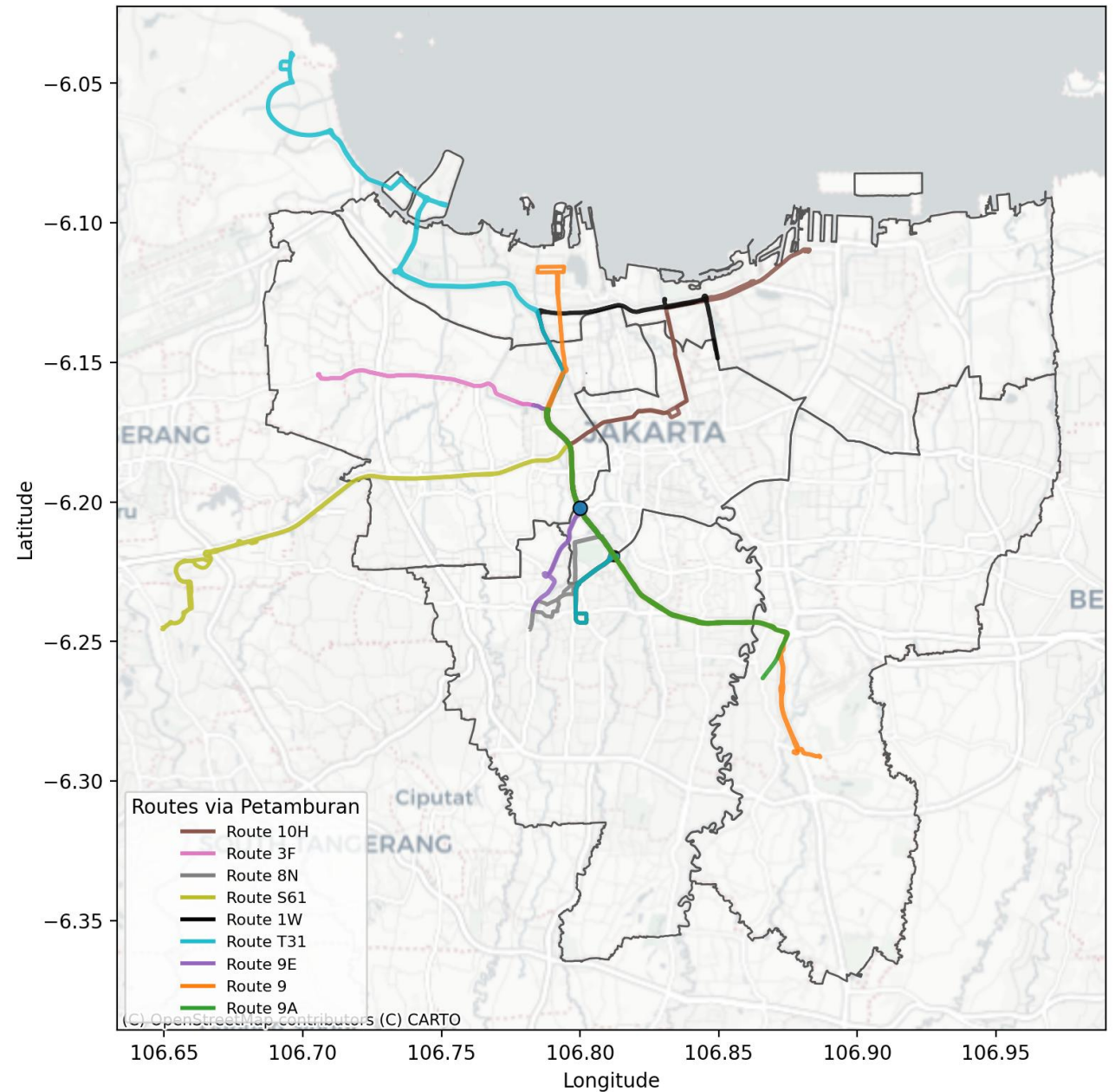
Galunggung – Camera Spot



Video Data Availability

Petamburan – Network Coverage

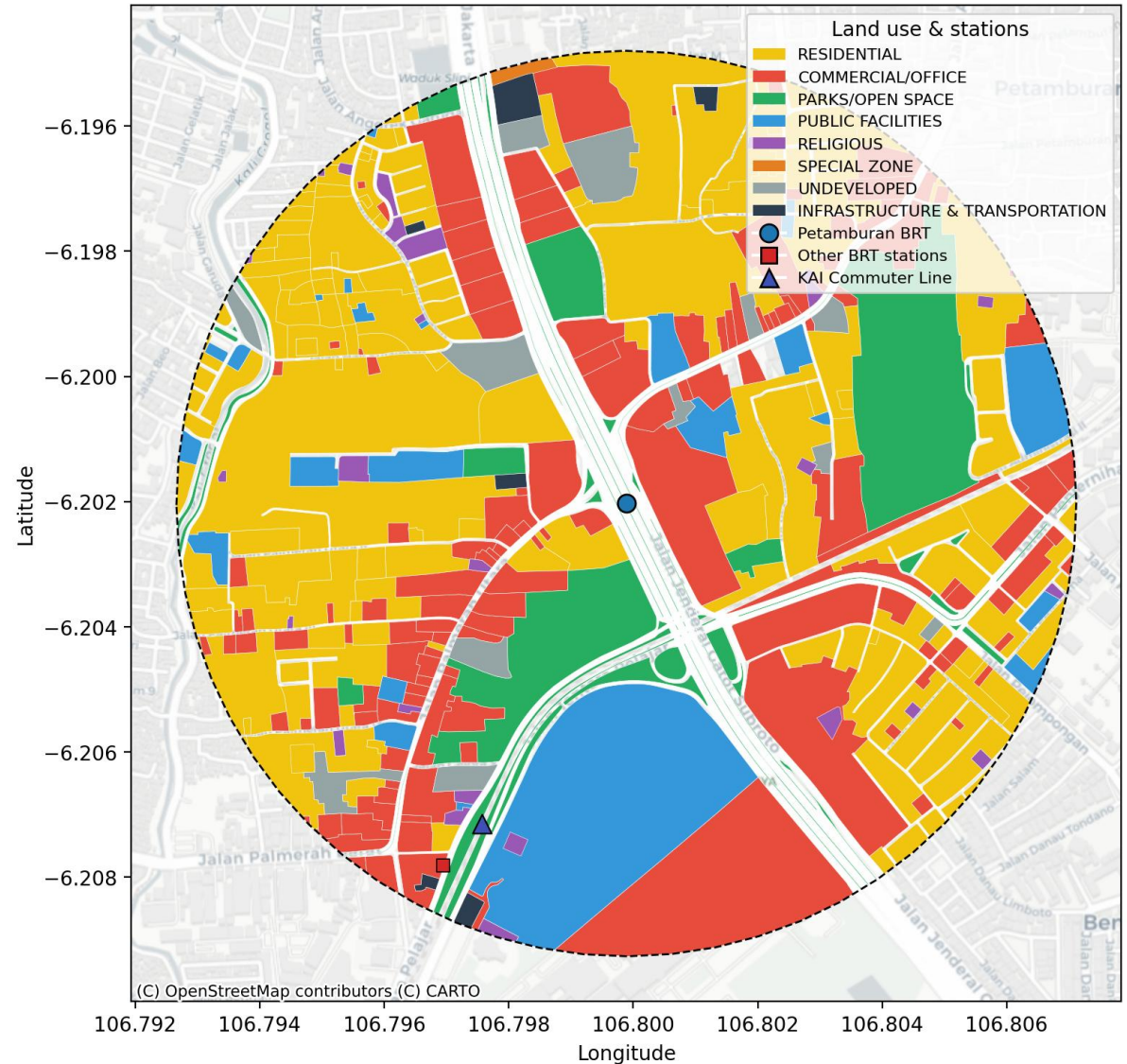
- Network coverage at Petamburan: a 9-route multi-directional hub, connecting coastal Tanjung Priok – international seaport in northern Jakarta – western Tangerang, and southeastern corridors toward Bekasi.



Video Data Availability

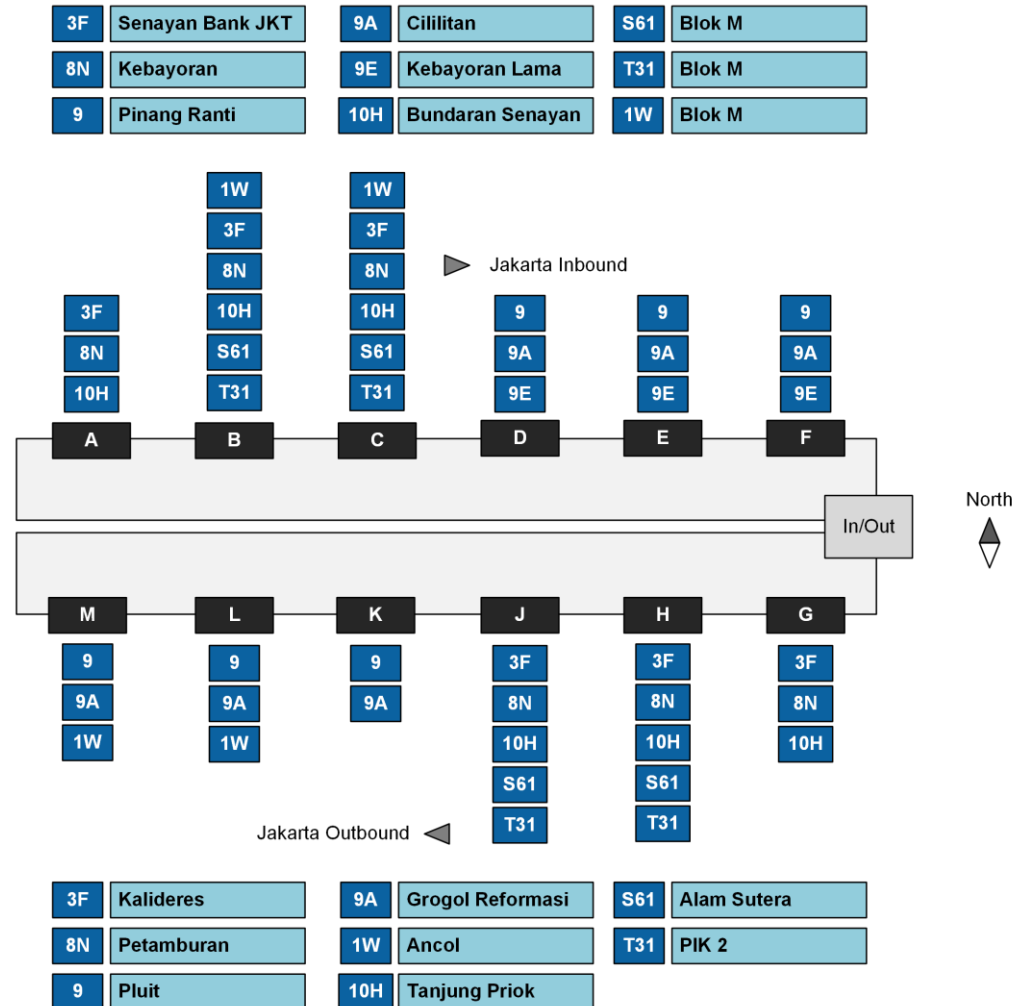
Petamburan – Land Use

- Petamburan Bus Station sits in dense residential-commercial fabric, while KAI Commuter Line and other BRT stops are located around 700m south, also provide the nearest transit gateway to the Indonesian National Parliament complex.



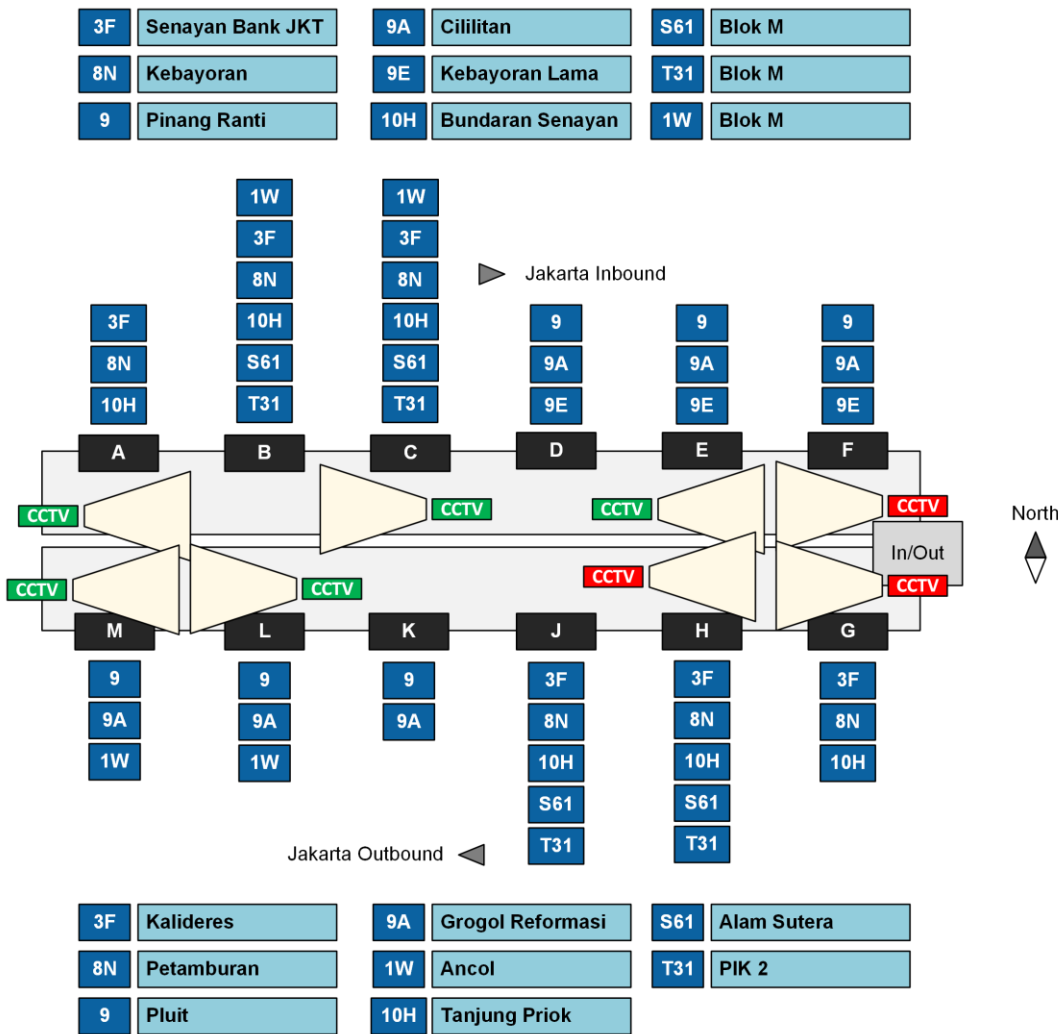
Video Data Availability

Petamburan – Station Layout



Video Data Availability

Petamburan – Camera Spot



Video Data Availability

Video Data – Period (Indonesian Western Time)

Cawang Sentral:

- Weekend (25 Oct 2025): 10:00 – 11:00 and 17:00 – 18:00 / **Saturday**
- Weekdays (27 Oct 2025): 07:00 – 08:00 (Peak AM); 13:00 – 14:00 (Off-Peak); and 17:30 – 18:30 (Peak PM) / **Monday**

Galunggung:

- Weekend (25 Oct 2025): 10:00 – 11:00 and 17:00 – 18:00 / **Saturday**
- Weekdays (27 Oct 2025): 07:00 – 08:00 (Peak AM); 13:00 – 14:00 (Off-Peak); and 17:30 – 18:30 (Peak PM) / **Monday**

Petamburan:

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Video Data Availability

Video Data Quality – Overview

Cawang Sentral:

- HJKL bays: minor obstruction only (ceiling fan, signage on H/L; J/K a bit far) — largely usable
- D/E/P/R bays: docking bay D is fully and consistently blocked by signage across every recording; E, P, R only partially obstructed
- "Priuk-facing" camera: **completely unusable** — docking bay cannot even be identified, obstructed visibility

Galunggung:

- All cameras rated Good quality; no fully-obstructed or unusable angle recorded
- Obstructions are localized and partial only (signage on B/L/C/K/M/A/H/E; ceiling fan on M/A) — most bays (A, D, E, G, H, J, K, L, C) come through fully clear
- Includes a dedicated passage camera (alighting ↔ boarding platform) rated Good/Clear/fully usable

Video Data Availability

Video Data Quality – Overview

Petamburan:

- Ranges from Good (K/L/M bays, fully clear) to Poor
- Poor-quality cameras show defects not seen at the other two stations: image blur with red/blue color distortion, a bay partially cropped out of frame, and severe frame ghosting/temporal duplication of subjects
- Common obstructions (ceiling fan, signage) also present on several bays (B, C, E, H)